Name: Ezaldin Nadam

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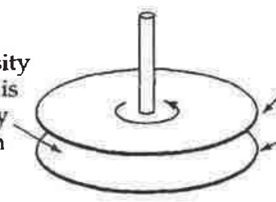
TRANSPORT PHENOMENA I, CHE 311

MIDTERM EXAM 1

CLOSED BOOK, TIME: 75 MINUTES

PARALLEL-DISK VISCOMETER

A fluid is placed in the gap between two horizontal circular disks of radius R. Disks are separated by a small distance B. The lower disk is held fixed. The upper disk is rotating at a constant angular velocity Ω.



Postulates:

- Newtonian fluid with constant density ρ and constant viscosity μ.
- $v_r = 0$ and $v_z = 0$.
- Pressure in the fluid is not a function of θ.
- Creeping flow assumption: The flow is very slow and the acceleration term $\frac{D[v]}{Dt}$ in the equation of motion is negligible.

20 1. Using the continuity equation, prove v_θ is not a function of θ at steady-state conditions.

10 2. Using the proper equation of motion at steady-state conditions, write down and simplify a partial differential equation for v_θ as a function of r and z.

15 3. Assuming v_θ may be written as r multiplied by a function of z,

$$v_\theta(r,z) = r\,f(z)$$

, convert the partial differential equation for v_θ to an ordinary differential equation for f.

20 4. Knowing $v_\theta(r, z = 0) = 0$ and $v_\theta(r, z = B) = r\,\Omega$, write down two boundary conditions for f, solve the ordinary differential equation derived in part 1.3. and evaluate the integration constants.

0 5. One can estimate the viscosity of the fluid by measuring the torque T_z required to rotate the upper disk at a constant angular velocity. Using the velocity profile derived in part 1.4., find an equation for the viscosity in terms of T_z, Ω, B and R. Hint: torque exerted by the liquid on unit area of the upper disk is the product of the radius r (lever arm) and viscous stress in θ-direction.

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1.) No p b/c it is a constant $\quad \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} = 0$

$$\frac{1}{r}\frac{\partial}{\partial r}(r v_r) + \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} = 0 \qquad \boxed{\frac{\partial v_\theta}{\partial \theta} = 0}$$ ✓

2.) Creeping flow

$$0 = -\frac{1}{r}\frac{\partial p}{\partial \theta} + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_\theta}{\partial r}\right) + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{1}{r^2}\frac{\partial^2 v_\theta}{\partial \theta^2} - \frac{v_\theta}{r^2}\right]$$?

$$\frac{\partial p}{\partial \theta} = 0 \qquad \frac{\partial v_\theta}{\partial \theta} \overset{\text{from 1.!}}{=} 0 \Rightarrow \frac{\partial^2 v_\theta}{\partial \theta^2} = 0$$

$$0 = \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_\theta}{\partial r}\right) + \frac{\partial^2 v_\theta}{\partial z^2} - \frac{v_\theta}{r^2}\right]$$?

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_\theta}{\partial r}\right) + \frac{\partial^2 v_\theta}{\partial z^2} - \frac{v_\theta}{r^2} = 0 \quad \text{divide by } \mu$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_\theta}{\partial r}\right) = \frac{\partial^2 v_\theta}{\partial r^2} + \frac{1}{r}\frac{\partial v_\theta}{\partial r}$$

$$\boxed{\frac{\partial^2 v_\theta}{\partial r^2} + \frac{1}{r}\frac{\partial v_\theta}{\partial r} - \frac{v_\theta}{r^2} + \frac{\partial^2 v_\theta}{\partial z^2} = 0}$$

3.) Assume $v_\theta(r,z) = r f(z)$ $\quad \frac{\partial v_\theta}{\partial r} = f(z) \quad \frac{\partial^2 v_\theta}{\partial r^2} = 0$

z-derivative

$$\frac{\partial v_\theta}{\partial z} = r f'(z)$$

$$\frac{\partial^2 v_\theta}{\partial z^2} = r f''(z)$$

$$0 + \frac{1}{r} f(z) - \frac{r f(z)}{r^2} + r f''(z) = 0$$

$$\frac{1}{r} f(z) - \frac{1}{r} f(z) = 0$$

$$r f''(z) = 0 \qquad f''(z) = 0$$

$$\boxed{\frac{d^2 f}{dz^2} = 0}$$ ✓

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4.)

$$V_\theta(r, z=0) = 0 \quad \& \quad V_\theta(r, z=B) = r\Omega$$

$r f(0) = 0 \rightarrow f(0) = 0$

$r f(B) = r\Omega \rightarrow f(B) = \Omega$

from R.3

$f''(z) = 0 \xrightarrow{\text{integrate}} f'(z) = C_1 \xrightarrow{\text{integrate}} f'(z) = C_1 z + C_2$

$f(0) = 0 \quad 0 = C_1(0) + C_2 \Rightarrow C_2 = 0$

$f(B) = \Omega \quad \Omega = C_1 B \Rightarrow C_1 = \frac{\Omega}{B} \qquad f(z) = \frac{\Omega}{B} z$

$$\boxed{V_\theta(r, z) = \frac{\Omega r z}{B}}$$